INCH-POUND
MIL-B-2678C
4 August 1989
SUPERSEDING
MIL-B-2678B
29 April 1966
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28 March 1963
(See 6.7 and 6.8)

MILITARY SPECIFICATION

BRUSHES, WIRE, BOILER TUBE, EXPANDING TYPE FOR USE WITH ELECTRIC AND AIR MOTOR DRIVEN BOILER TUBE CLEANING OUTFITS, AND FIXED SECTIONAL HAND TYPE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers expanding type of wire brushes for use with boiler tube cleaning outfits, and fixed sectional type of wire brushes for hand cleaning interior walls of boiler tubes.

1.2 Classification. Brushes shall be of the following types and classes, as specified (see 6.2).

Type I - Expanding type.
- Class A - For use with electric-motor-driven boiler tube cleaning outfits (2 inches or smaller) (Army and Air Force use only).
- Class B - For use with air-motor-driven boiler tube cleaning outfits (2 inches or smaller).
- Class C - For use with either electric- or air-motor-driven boiler tube cleaning outfits (with universal joint).
- Class D - For use with either electric- or air-motor-driven boiler tube cleaning outfits (larger than 2 inches).

Type II - Fixed sectional hand type for use with rigid rods.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-P-40 - Preservation and Packing of Hand Tools; Tools and Tools Accessories for Power Driven, Metal and Woodworking Machinery.

MILITARY

MIL-T-17370 - Tube and Flue Cleaner, Boiler, Internal, Air-Motor-Driven.
MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-151 - Metals; Test Methods.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Type I, classes B, C, and D brushes furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.3 and 6.4).

3.2 First article. When specified (see 6.2), a sample of type I, class A, and type II brush shall be subjected to first article inspection (see 6.5) in accordance with 4.4.

3.3 Materials. Materials to be used for brushes shall be as specified herein.

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3.4 Type I brushes. Brushes shall expand and contract to provide automatic adjustment of the outside diameter of brushes to accommodate variations in the inside diameter of straight and curved sections of boiler tubes.

3.4.1 Dimensions. Dimensions of brushes and the degree of expansion and contraction provided shall be such that, when operated with the corresponding type and size of an approved make of boiler-tube cleaning outfit, the brushes shall pass through and effectively operate in straight and curved sections of sizes of boiler tubes shown in table I (see 6.2).

TABLE I. Dimensions of curved boiler tube sections.

Size of brush and outside diameter of boiler tube	Arc radius	Arc length	Maximum crush of tube
(inches)	(inches)	(degrees)	(inch)
1	5	75	0.050
1-1/8	5	75	.050
1-1/4	5	75	.062
1-3/8	6	75	.062
1-1/2	6	75	.075
1-5/8	6	75	.075
1-3/4	6	75	.100
2	6	75	.120
2-1/2	12	75	.140
2-3/4	12	75	.165
3 to 4	12	75	.190
4-1/2	14	75	.215
5	16	75	.240

3.4.2 Hardness. All parts of brushes shall be made of steel. Working parts exposed to wear shall not exceed a maximum hardness of Rockwell B-68 or a minimum of Rockwell B-60.

3.4.3 Bodies of brushes. Bodies of brushes shall conform to the dimensions shown on figure 1 so that refills may be interchangeable regardless of their manufacturer.

3.4.4 Refills. Refills shall be supplied in sets, each of which shall comprise the parts necessary to renew one complete brush (body, end caps, and screw shall not be included).

3.4.5 Brush replacement. Brushes shall be constructed so that the bristle sections can be replaced without renewing the brush body.

3.4.6 Bristles. Only steel wire bristles shall be used in brushes intended for removing deposits on the interior of boiler tubes.

3.4.6.1 Wire material. Bristles shall be of bright, tinned, high-tempered steel wire and shall meet the chemical composition requirements shown in table II.

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TABLE II. Chemical composition requirements.

Carbon (percent)	Manganese (percent)	Phosphorous (percent)	Sulfur (percent)	Silicon (percent)
0.65 to 0.75	0.50 to 0.80	0 to 0.04	0 to 0.050	0.12 to 0.30

3.4.6.2 Strength. The wire shall meet the following tensile strength requirement and upper elongation limit:

Tensile strength - 250,000 to 300,000 pounds per square inch.
Elongation - 2 percent in test wire of 10 inches or greater length.

3.4.6.3 Diameters. Bristles shall be 0.0085 to 0.0095 inch in diameter in brushes of size 2 inches or smaller, and 0.016 to 0.020 inch in diameter in brushes of all larger sizes.

3.4.6.4 End finishing. End of bristles shall not be trimmed by grinding.

3.4.7 Threaded connections. Bodies of brushes shall be threaded for connection with flexible brush holders that conform to the requirements shown in table III.

TABLE III. Threading of bodies of brushes for connection with flexible brush holders.

Class of brush	Size of brush (inches)	Size of thread (inch)	Threads per inch (number)	Length of threaded section (inch)	Male or Female
A, B, and C	1 to 1-1/2	5/16	18	5/16	Male
A, B, and C	1-5/8 to 1-3/4	7/16	14	3/8	Male
A	2	9/16	12	1/2	Female
B and C	2	7/16	14	3/8	Male
D	2-1/2 to 4	9/16	12	1/2	Female
D	4-1/2 to 5	7/8	9	9/16	Female

3.4.7.1 Threads. Thread dimensions, tolerances, and gauges for determination of interchangeability, shall conform to FED-STD-H28 and the requirements shown in table IV. Unified coarse thread (UNC) series 2A-2B shall be the standard for screw threading; UNC series 3A-3B standards are also acceptable.

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TABLE IV. Dimensions.

Thread size	Major diameter		Pitch diameter		Minor diameter (max) (inch)
	Maximum (inch)	Minimum (inch)	Maximum (inch)	Minimum (inch)	
5/16 by 18	0.3125	0.3043	0.2764	0.2734	0.2443
7/16 by 14	.4375	.4277	.3911	.3875	.3499
9/16 by 12	.5625	.5513	.5084	.5044	.4603
7/8 by 9	.8750	.8610	.8028	.7979	.7387

3.4.8 Brush dimensions.

3.4.8.1 Class A. Brushes of size 2 inches or smaller for use with MIL-T-17370 electric-motor-driven outfits for Army and Air Force use only, shall conform to the dimensions shown in table V. Each brush shall be provided with mechanical means, independent of and in addition to the centrifugal force caused by rotation of the brush, for automatic adjustment of the outside diameter of the brush to variations in the inside diameter of the tubes.

TABLE V. Dimensions of class A brushes (sizes 2 inches and smaller).

Outside diameter of tube (inches)	Wall thickness of tube (inch)	Inside diameter of tube (inches)	Diameter of brush		Free length of bristle (min) (inch)	Wire ends per brush (min) (number)
			Expanded (min) (inches)	Contracted (max) (inches)		
1	0.085 to 0.095	0.810 to 0.830	0.820	0.733	0.120	3360
1	0.109	0.782	.792	.705	.106	3360
1-1/8	.109	.907	.917	.830	.169	3360
1-1/8	.120	.885	.895	.808	.158	3360
1-1/4	.109	1.032	1.042	.920	.178	4800
1-1/4	.134	0.982	0.992	.870	.153	4800
1-3/8	.120	1.135	1.145	1.022	.229	4800
1-3/8	.134	1.107	1.117	0.994	.215	4800
1-1/2	.120	1.260	1.270	1.110	.243	5760
1-1/2	.134	1.232	1.242	1.082	.229	5760
1-5/8	.134	1.357	1.367	1.172	.227	5760
1-3/4	.134	1.482	1.492	1.298	.290	5760
1-3/4	.156	1.438	1.448	1.254	.268	5760
2	.134	1.732	1.742	1.542	.287	8960
2	.165	1.670	1.680	1.480	.257	8960

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3.4.8.2 Class B. Brushes of sizes 2 inches or smaller for use with MIL-T-17370 air-motor-driven outfits shall conform to the dimensions shown in table VI.

TABLE VI. Dimensions of class B brushes (sizes 2 inches and smaller).

Outside diameter of tube (inches)	Wall thickness of tube (inch)	Inside diameter of tube (inches)	Diameter of brush		Free length of bristle (min) (inch)	Wire ends per brush (min) (number)
			Expanded (min) (inches)	Contracted (max) (inches)		
1	0.085 to 0.095	0.810 to 0.830	0.830	0.689	0.118	2560
1	0.109	0.782	.802	.66	.104	2560
1	.134	.732	.751	.610	.076	2560
1-1/8	.109	.907	.927	.786	.167	2560
1-1/8	.120	.885	.905	.764	.156	2560
1-1/4	.109	1.032	1.052	.894	.198	2560
1-1/4	.120	0.010	1.030	.872	.186	2560
1-1/4	.134	0.982	1.002	.844	.173	2560
1-3/8	.120	1.135	1.155	.996	.249	2560
1-3/8	.134	1.107	1.127	.968	.235	2560
1-1/2	.120	1.260	1.280	1.120	.248	4800
1-1/2	.134	1.232	1.252	1.092	.234	4800
1-5/8	.134	1.357	1.377	1.176	.232	5760
1-3/4	.134	1.482	1.502	1.302	.295	5760
1-3/4	.156	1.438	1.458	1.258	.273	5760
2	.134	1.732	1.752	1.522	.292	8960
2	.148	1.704	1.724	1.494	.277	8960
2	.165	1.670	1.690	1.460	.262	8960
2	.180	1.640	1.660	1.430	.247	8960

3.4.8.3 Class C. Brushes of sizes 2 inches or smaller with universal joint for use with either electric-motor-driven or air-motor-driven outfits shall conform to the dimensions shown in table VII. The universal joint of a type satisfactory to the contracting activity shall be attached to the end of the brush that joins the brush holder, to permit operation in tubes bent to a radius of 2-3/4 inches through 180 degrees where the crush of tube does not exceed 1.5 times that shown in table I.

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TABLE VII. Dimensions of class C brushes (sizes 2 inches and smaller).

Outside diameter of tube (inches)	Wall thickness of tube (inch)	Inside diameter of tube (inches)	Diameter of brush		Free length of bristle (min) (inch)	Wire ends per brush (min) (number)
			Expanded (min) (inches)	Contracted (max) (inches)		
1	0.085 to 0.095	0.810 to 0.830	0.810	0.705	0.125	1920
1	0.109	0.782	.782	.677	.110	1920
1	.134	.732	.732	.627	.080	1920
1-1/8	.109	.907	.907	.802	.173	1920
1-1/8	.120	.885	.885	.780	.162	1920
1-1/4	.109	1.032	1.042	.956	.163	1920
1-1/4	.134	0.982	0.992	.906	.138	1920
1-3/8	.120	1.135	1.145	1.059	.214	1920
1-3/8	.134	1.107	1.117	0.031	.200	1920
1-1/2	.120	1.260	1.270	1.070	.220	3840
1-1/2	.134	1.232	1.242	1.042	.206	3840
1-5/8	.134	1.357	1.367	1.125	.205	3840
1-3/4	.134	1.482	1.492	1.246	.240	3840
1-3/4	.156	1.438	1.448	1.200	.220	3840
2	.134	1.732	1.742	1.422	.321	6400
2	.148	1.704	1.714	1.394	.305	6400
2	.165	1.670	1.680	1.360	.290	6400

3.4.8.4 Class D. Brushes of sizes larger than 2 inches, up to and including 5 inches, for use with electric-motor-driven or air-motor-driven outfits, shall conform to the dimensions shown in table VIII.

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TABLE VIII. Dimensions of class D brushes (sizes larger than 2 inches, up to and including 5 inches).

Outside diameter of tube (inches)	Wall thickness of tube (inch)	Inside diameter of tube (inches)	Diameter of brush		Free length of bristle (min) (inch)	Wire ends per brush (min) (number)
			Expanded (min) (inches)	Contracted (max) (inches)		
2-1/2	0.120	2.260	2.369	1.924	0.307	1800
2-1/2	.134	2.232	2.341	1.896	.293	1800
2-1/2	.148	2.204	2.313	1.868	.279	1800
2-1/2	.165	2.170	2.279	1.834	.262	1800
2-3/4	.120	2.510	2.519	2.174	.432	1800
2-3/4	.134	2.482	2.591	2.146	.418	1800
2-3/4	.148	2.454	2.563	2.118	.404	1800
2-3/4	.165	2.420	2.529	2.084	.387	1800
3	.134	2.732	2.825	2.417	.391	2520
3	.165	2.670	2.763	2.355	.360	2520
3	.250	2.500	2.667	2.259	.312	2520
3-1/4	.134	2.982	3.075	2.667	.516	2520
3-1/4	.165	2.920	3.013	2.605	.485	2520
3-1/4	.250	2.750	2.843	2.435	.400	2520
3-1/2	.120	3.260	3.353	2.902	.428	2520
3-1/2	.134	3.232	3.325	2.874	.412	2520
3-1/2	.165	3.170	3.263	2.812	.381	2520
3-1/2	.180	3.140	3.233	2.782	.365	2520
4	.134	3.732	3.825	3.374	.662	2520
4	.165	3.670	3.763	3.312	.631	2520
4	.203	3.594	3.687	3.236	.593	2520
4	.250	3.500	3.593	3.142	.546	2520
4-1/2	.180	4.140	4.233	3.729	.578	2520
4-1/2	.203	4.094	4.187	3.683	.555	2520
4-1/2	.220	4.060	4.153	3.649	.538	2520
4-1/2	.250	4.000	4.093	3.589	.508	2520
4-1/2	.280	3.940	4.033	3.529	.478	2520
4-1/2	.293	3.914	4.007	3.503	.465	2520
4-1/2	.343	3.814	3.907	3.403	.415	2520
5	.300	4.400	4.493	3.989	.415	2520

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3.5 Type II brushes. Type II hand brushes shall be used only with rigid rods; they shall not be used with motor-driven boiler tube cleaning outfits.

3.5.1 Construction. Sectional brushes shall be made up of tubular sections; each section shall consist of steel wire fagots mounted on a tubular steel or brass drum. The drum of each section shall be mounted on a steel spindle that has threaded end fittings for joining sections securely together. The assembled brush shall be similar to the one shown on figure 2.

3.5.2 Fagots. Each fagot shall consist of approximately 40 hardened and polished steel wires, 0.009 inch in diameter. The fagots shall be doubled and securely fastened at the middle, through a hole in the drum, by a steel or bronze binding wire, or by stapling. Fagot ends shall be cut evenly and to such length as to produce a brush of the required outside diameter. Fagots shall be arranged in four equidistant rows around the circumference of each drum; the rows shall be staggered so that the surface is uniformly covered.

3.5.3 Bearing collars. Each drum shall have a bearing collar fitted loosely over the spindle to prevent wear on the binding wires.

3.5.4 End fittings. The threaded end fittings shall be machined from hexagon steel or brass rod, with sufficient length left unturned to permit gripping with a wrench. The end clamp collars or washers for brushes shall be conical or flat in shape and made of steel or brass. The outer end fittings shall be rounded at the end of the hexagonal section. The inner end fitting shall be tapped 3/8-inch nominal pipe size, at the end of the hexagonal section, in accordance with FED-STD-H28. The fittings shall be mounted on spindles, threaded on each end to UNC-2A fit or tighter, and of sufficient length to permit clamping the sections of the brush together. Spindle sizes shall conform to those listed in table IX.

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TABLE IX. Dimensions of type II brushes.

Size (outside diameter) of tube (inches)	Thickness of wall (mils)	Diameter over fagots 1/ (inches)	Outside diameter of drum 2/ (inches)	Width of drum (inches)	Inside diameter of collar (max) (inch)	Thickness of collar (inch)	Size of spindle (max) (inches)	Number of fagots per section
2	95	1-27/32	1	1-1/8	1/2	1/16	3/8 by 6	40
2	134	1-49/64	1	1-1/8	1/2	1/16	3/8 by 6	40
2	165	1-45/64	1	1-1/8	1/2	1/16	3/8 by 6	40
2-1/2	109	2-5/16	1-3/8	1-1/8	9/16	1/16	7/16 by 6-1/4	60
3	109	2-13/16	1-3/4	1-1/4	5/8	1/4	1/2 by 5-1/2	80
3	120	2-51/64	1-3/4	1-1/4	5/8	1/4	1/2 by 5-1/2	80
3	165	2-45/64	1-3/4	1-1/4	5/8	1/4	1/2 by 5-1/2	80
3-1/2	120	3-19/64	2-1/8	1-1/4	11/16	1/4	9/16 by 7-3/4	100
4	134	3-49/64	2-1/2	1-1/4	3/4	1/4	5/8 by 7-3/4	120
4	165	3-45/64	2-1/2	1-1/4	3/4	1/4	5/8 by 7-3/4	120
4	203	3-5/8	2-1/2	1-1/4	3/4	1/4	5/8 by 7-3/4	120

1/ A minus tolerance of 1/32 inch in diameter over fagots will be permitted.
2/ Measured parallel to axis of drum.

3.5.5 Sections per brush. Brushes shall have four sections for 2 and 2-1/2 inch tubes and shall have three sections for larger tubes.

3.5.6 Dimensions. Dimensions of brushes and brush sections shall be as shown in table IX.

3.5.7 Brush sections (refills). When ordered separately, brush sections shall conform to the requirements of 3.5.2 and 3.5.6. Brush sections (refills) shall be similar to those shown on figure 3.

3.5.8 Marking. The size of tube (outside diameter and thickness of wall) shall be stamped on each brush section.

3.6 Workmanship. Nonconformance to requirements of this specification shall be cause for rejection.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) First article inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).

4.3 Qualification inspection (for type I, class B, C and D brushes only). Qualification inspection shall be conducted at a laboratory satisfactory to NAVSEA. Qualification inspection shall consist of the examinations of 4.6.1 through 4.6.1.3 and the tests of 4.8.2 and 4.8.3. Tests shall be performed on each type, class, and size of brush for which qualification approval is desired (see 6.4).

4.4 First article inspection and testing of each size of type I, class A and type II brush. Each first article sample (see 3.2) shall be visually and dimensionally examined and tested as specified in 4.8.1 and 4.8.3 (see 6.3).

4.5 Quality conformance inspection. A quality conformance inspection shall consist of examinations and test as specified in 4.6.1 through 4.6.1.3 (see 6.3).

4.5.1 Sampling for quality conformance inspection. The brushes or refill sets of one class and size offered for delivery at one time shall be considered a single lot for purposes of quality conformance inspection.

4.5.2 Sampling for visual examination. A random sample of brushes or refill sets shall be selected from each lot. Each of the sample units shall be subjected to the visual examination specified in 4.6.1.

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4.5.3 Sampling for dimensional examination. A random sample of brushes or refill sets shall be selected from each lot. Each of the sample units shall be subjected to the dimensional examination specified in 4.6.1.1.

4.5.4 Sampling for bristle count. A random sample of brushes or refill sets shall be selected from each lot and shall be subjected to the examination specified in 4.6.1.2.

4.6 Inspection.

4.6.1 Visual examination. Each of the sample brushes or refill sets shall be subjected to visual examination (other than dimensional examination and bristle count) to determine conformance to this specification.

4.6.1.1 Dimensional examination. Each of the samples shall be subjected to a careful dimensional check to determine conformance to the applicable dimensional requirements of 3.4.4, 3.4.7.3, 3.5.2, and tables III to IX, inclusive. In checking the free length and diameter of the bristle, this measurement may be made on a single bristle cartridge from each sample unless the inspector considers further examination desirable.

4.6.1.2 Bristle count. The number of wire ends in one fagot of each of the sample brushes selected in accordance with 4.5.4 shall be carefully counted. The fagot to be counted shall be selected from a different row in each of the sample brushes or refill sets. The number of wire ends per brush shall then be calculated by multiplying the count of wire ends by number of fagots in the corresponding brush. If the resulting number is not in conformance to the requirements of tables V to IX inclusive, the entire lot shall be rejected.

4.6.1.3 Defects for visual and dimensional examination and bristle count. Defects shall be in accordance with table X.

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TABLE X. Classification of defects.

Categories	Defects	Visual	Dimensional	Bristle count	Sampling para.
Critical:					
1	None defined				
Major:					
101	Evidence of use of unauthorized material.	X			4.5.2
102	Brushes not class specified; not expanding and contracting type.	X			4.5.2
103	Brush body dimensions and configuration not in conformance with approved figure; not within tolerances specified.				4.5.3
104	Bristles not bright or tinned.	X			4.5.2
105	Brush diameter not within tolerance specified.	X			4.5.3
106	Evidence of brushes being trimmed by grinding.	X			4.5.2
107	Brush diameter (expanded and contracted), free length of bristles, and number of wire ends per brush nonconforming (for type I brushes).	X	X		4.5.3 4.5.4
108	Bodies of brush not of thread size or threads per inch specified, length of threaded section nonconforming.	X			4.5.3
109	Threads not of class or fit specified; major, minor, and fitted diameter not within tolerance specified.		X		4.5.3
110	Universal joint nonconforming (for class A and B universal brushes), not attached as specified.	X			

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TABLE X. Classification of defects - Continued.

Categories	Defects	Visual	Dimen-sional	Bristle count	Sampling para.
111	Bristle sections cannot be renewed without re-newing body (for type I brushes).	X			4.5.2
112	Refill sets not complete; units missing (if required).	X			4.5.2
113	Brushes do not comply with approved drawings.	X			4.5.2 4.5.3
114	Workmanship poor.	X			4.5.2
115	Automatic adjustment of outside of brushes not operatable (for type I brushes).	X			4.5.2
116	Parts of refill sets not correct count (for type I brushes).	X			4.5.2
Minor:					
201	None defined.				

4.7 Comparison testing. One sample complete brush and wire shall be selected and forwarded to a laboratory satisfactory to the contracting activity for comparison test. The sample shall be subjected to any test considered necessary to determine that the sample is equal to the sample upon which qualification approval was based. If an unsatisfactory report is obtained, the contracting activity shall be notified immediately.

4.8 Tests.

4.8.1 First article test for type I, class A, and type II brushes only. Each first article sample (see 4.4) shall be fitted or assembled into a complete electric-motor-driven tube and flue cleaner for the sample brush or refill set (cleaner shall conform to MIL-T-17370). The brush shall be able to pass through and effectively operate in straight and curved sections of sizes of boiler tubes for which designated. This sample shall be operated in cleaning of dirty boiler tubes for a period of not less than 1 hour. After the completion of the test, both the sample and the tubes involved shall be inspected. The sample shall not show wear exceeding 10 percent of total original bristle length and the tubes shall show clean metal and no damage such as scratches, grooving or dents caused by the sample cleaning.

4.8.2 Qualification test for type I, class B, C, and D brushes only. Each sample selected in accordance with 4.5.1 shall be assembled or fitted into a standard cleaning unit selected from stock suitable for brush or refill set size. The brush shall be able to pass through and effectively operate in straight and

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curved sections of sizes of boiler tubes for which designated. This sample shall be operated in cleaning of dirty boiler tubes for 5 hours. Both the sample and the tubes involved shall then be inspected to determine the acceptability of the sample as compared to the condition and action of qualified standard stock brushes after similar operation for 5 hours.

4.8.3 Wire. Specimens of the bristle wire for brushes of each size group as specified in 3.4.7.3 shall be tested and analyzed to determine conformance to 3.4.7. Physical tests shall be conducted in accordance with the applicable requirements of FED-STD-151.

4.9 Inspection of packaging. Sample packs, and the inspection of the preservation, packing, and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. Packaging.

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Packaging requirements. Brushes and brush sections shall be preserved level A or Commercial, packed level A, B, or Commercial as specified (see 6.2) and marked in accordance with PPP-P-40 and shall include barcodes and applicable packaging acquisition options therein as specified (see 6.2). Brushes shall be packaged by units, section refills by sets sufficient to renew one brush. In addition, for Navy acquisitions, the following applies:

(a) Navy shipboard stowage fire-retardant requirements.

1. Treated lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B	- Type II - weather resistant. Category 1 - general use.
Level C	- Type I - non-weather resistant. Category 1 - general use.

2. Fiberboard. Unless otherwise specified (see 6.2), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard and cleated fiberboard boxes including interior packaging forms shall meet the flamespread and the specific optic density requirements of PPP-F-320 and amendments thereto.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Brushes and brush sections covered by this specification are intended for use in cleaning the internal surfaces of boiler tubes, condenser tubes, and heat exchanger tubes.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
(d) Dimensions of curved sections (see 3.4.1).
(e) Dimensions of brushes (see tables V to IX inclusive).
(f) When fire-retardant treated materials are not required (see 5.1).
(g) Levels of preservation, packing, and marking required (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjuction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.4	DI-T-4902	First article inspection report	----
4.5	DI-T-5329	Inspection and test report	----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 2678 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded

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contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests must be made in accordance with "Provisions Governing Qualification SD-6" (see 6.4.1).

6.4.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first lot production items, a standard production item from the contractor's current inventory (see 3.2), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Subject term (key word listing).

Bearing collars
Bristles
End fittings
Fagots
Spindles
Threaded connections

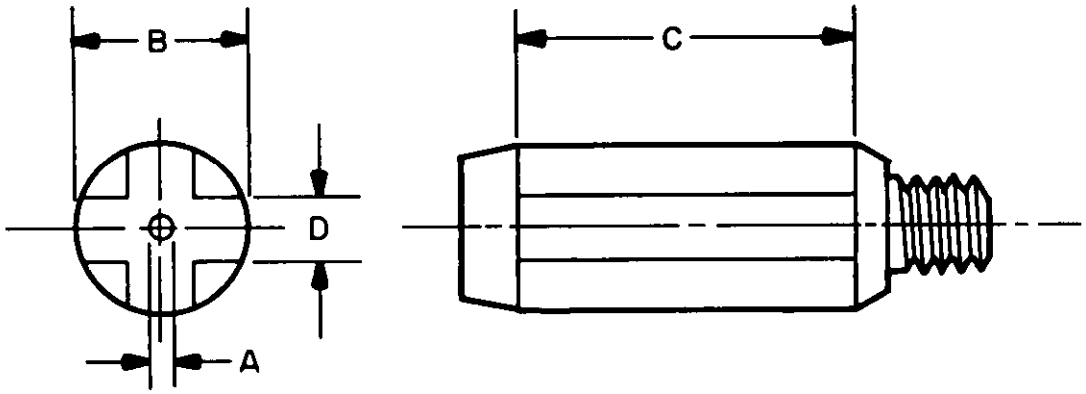
6.7 Supersession data. This revision of MIL-B-2678 includes the requirements of MIL-B-2679 for fixed sectional hand type of boiler tube cleaning brushes.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - GL
Navy - SH
Air Force - 99

Preparing activity:
Navy - SH
(Project 5120-C027)

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Dimensions of class A - Sizes up to and including 2 inches.

Brush size (inches)	A (inch)	B (inches)	C (inches) (±0.005)	D (inch) (+0.003 - .000)	Number of slots
1 and 1-1/8	3/32	9/16	1-15/16	0.159	6
1-1/4 and 1-3/8	3/32	11/16	1-11/16	.222	
1-1/2	3/32	13/16	1-13/32	.222	6
1-5/8 and 1-3/4	1/8	15/16	1-27/64	.224	6
2	3/16	1-3/16	1-5/8	.224	8

Dimensions of class B (sizes up to and including 2 inches)

Brush size (inches)	A (inch)	B (inches)	C (inches) (±0.005)	D (inch) (+0.003 - .000)	Number of slots
1 and 1-1/8	1/16	9/16	63/64	0.222	4
1-1/4 and 1-3/8	5/64	11/16	63/64	.222	4
1-1/2	11/64	13/16	1-7/32	.222	6
1-5/8 and 1-3/4	3/16	15/16	1-27/64	.224	6
2	3/8	1-3/16	1-5/8	.224	8

SH 538

FIGURE 1. Boiler tube expanding type body.

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Dimensions of class C (sizes up to and including 2 inches)

Brush size (inches)	A (inch)	B (inches)	C (inches) (±0.005)	D (inch) (+0.003 - .000)	Number of slots
1 and 1-1/8	1/16	9/16	51/64	0.222	4
1-1/4 and 1-3/8	9/64	3/4	51/64	.222	4
1-1/2	3/16	7/8	63/64	.222	6
1-5/8 and 1-3/4	1/4	1	63/64	.222	6
2	9/32	1-1/8	1-11/64	.222	8

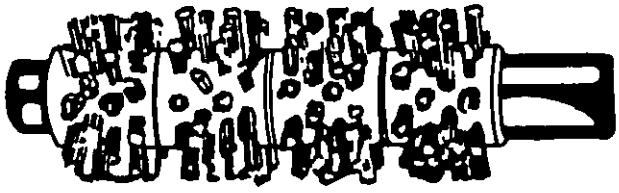
Dimensions of class D (sizes larger than 2 inches, through size 5 inches)

Brush size (inches)	A (inch)	B (inches)	C (inches) (±0.005)	D (inch) (+0.005 - .000)	Number of slots
2-1/2	1/2	1-9/16	1-51/64	0.442	6
3 and 3-1/4	5/8	2	2-3/32	.570	6
3-1/2 and 4	3/4	2-3/8	2-3/32	.570	6
4 and 4-1/2	1-9/16	3	2-1/8	.570	6
5	1-9/32	3-3/8	2-1/8	.570	6

SH 538

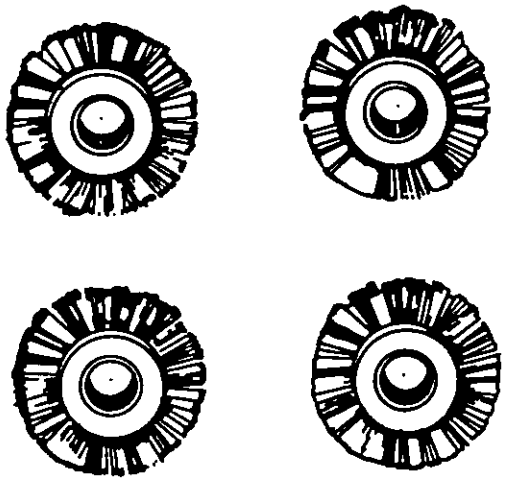
FIGURE 1. Boiler tube expanding type body - Continued.

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SH 13231772

FIGURE 2. Assembled boiler tube brush.



SH 13231773

FIGURE 3. Brush sections (refills).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-2678C	BRUSHES, WIRE, BOILER TUBE, EXPANDING TYPE FOR USE WITH ELECTRIC AND AIR MOTOR DRIVEN BOILER TUBE CLEANING OUTFITS, AND FIXED SECTIONAL HAND TYPE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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